Chemigenetic Tandem-Tetracysteine-Enhanced BRET Sensors for Protease Activity Quantification in Plasma and Drug Screening

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Abstract

Proteases play a crucial role in essential physiological processes and are recognized as therapeutic targets and diagnostic biomarkers for different pathologies. Here, we developed a new class of chemigenetic Tetracysteine-based Protease Activity Reporters (TePARs) by sandwiching peptides containing protease cleavage sites in between the NanoLuc protein and 12 amino acid tetracysteine motif conjugated to synthetic fluorophores. TePARs used bioluminescence resonance energy transfer (BRET) to shift emitted light from green or red to blue upon proteolysis, allowing detection with either a plate reader or a simple smartphone camera. With further optimization, TePAR 2.0 utilized tandem tetracysteine peptides to achieve unprecedented BRET efficiency (~90%), large dynamic range (>40-fold), and a picomolar limit of detection (LoD). Moreover, TePARs feature a modular design that enables the generation of a wide range of protease sensors by simply swapping peptide substrates for different proteases while maintaining exceptional performance. We used TePAR to quantify active proteases at physiologically relevant concentrations in challenging biological samples, yielding results consistent with those from standard testing methods. Notably, TePARs enable the quantification of MMP-9 plasma levels of colorectal cancer patients at the point of care. TePAR assay distinguished patients with colorectal cancer, including those with early-stage adenomatous polyps, from healthy controls with accuracies of over 90%, thus expanding their utility for point-of-care diagnostics. Furthermore, we also demonstrate that TePAR enables camera-based drug screening and characterization without the need for specialized instruments.

Introduction

Aberrant protease activity drives pathogenesis of multiple diseases, including colorectal and breast cancer[1], cardiovascular disease such as atherosclerosis and hypertension[2], neurodegenerative disorders such as Alzheimer's and Parkinson's diseases[3], and inflammation such as dry eye syndrome[4], inflammatory bowel diseases (IBD) and chronic obstructive pulmonary diseases (COPD)[5]. Given their substantial participation in physiological and pathological processes, along with their intricate regulatory mechanisms, proteases are widely recognized as both potential drug targets and valuable diagnostic biomarkers. As a result, there is intense interest in developing effective detection methods for diagnostics and drug discovery[6].

In typical protease detection efforts, protease detection assays could be grouped into affinity and activity assays. Affinity methods such as Western Blot and ELISA (enzyme-linked immunosorbent assay) depend on the affinity and specificity of antibody-antigen interactions[7]. The immunoassays are the gold standard for the detection and quantification of protein biomarkers. Such assays can detect proteins of interest at the pg/mL level and are compatible with different samples in a complex mixture. But they are expensive and time-consuming, which limits its use in rapid clinical diagnosis. Since affinity assays detect protease regardless of activity, activity assays are more applicable for functional protease detection. Activity assays, such as fluorescence resonance energy transfer (FRET) assays[8] and intensity-based turn-on assays[9–12], are commonly used for protease detection in cellular environments. However, their applicability in point-of-care settings is limited due to the requirement for excitation light.

Over the past few years, there has been an upsurge in the use of bioluminescent sensor proteins for point-of-care diagnostics due to the superior NanoLuc (NLuc) platform[13] and its flexibility in multiple point-of-care assay formats. Examples include intensity-based luminescent biosensor like BAT for SARS-CoV-2 spike protein[14] and FlipNanoLuc for coronavirus 3CL-protease[15]. Additionally, NLuc, a blue light-emitting donor, has also thrived in ratiometric sensor design. The Johnsson Lab has pioneered a series of novel bioluminescence resonance energy transfer (BRET) sensor for metabolite quantification using Snifit concept[16–19]. There are also activity-based BRET sensors for measuring intracellular protease activity[20,21] and in vitro assays[22].

However, previous bioluminescent sensors for protease activity assays have relied on fluorescent proteins (FPs), which present three limitations. First, all FPs are relatively large (~27 kDa), leading to a strong steric hindrance effect that slows down the proteolysis reaction. Second, their large size also hampers energy transfer efficiency, especially when compared to FRET-based small-molecule probes. Despite attempts to enhance BRET efficiency through linker engineering[23] or dual acceptor FPs strategies [24], the improvements have been minimal and challenging. Finally, these sensors exhibit limited brightness, especially when using red fluorescent proteins as acceptors. This limitation renders them unsuitable for camera-based assays that rely on the blue and red color channels of standard complementary metal-oxide-semiconductor (CMOS) and charge-coupled device (CCD) image sensor. Besides developing single-protease specific indicator, having a toolkit of indicators for all proteases would be especially useful. However, it is challenging to generalize the concept by simply swapping the protease-responsive specific peptide, and optimizing each protease sensor individually is a laborious task.

Here, we designed a novel class of Tetracysteine-based Protease Activity Reporters (TePARs). The sensor is a fusion protein of tetracysteine tag[25,26], the luciferase NLuc and the specific substrate peptide for protease cleavage (Fig. 1a). The fluorogenic biarsenical (Fig. 1b, c) is covalently labeled to the sensor protein through the self-labeling tetracysteine tag[25]. In the presence of the protease analyte, the sensor protein is cleaved, separating the fluorophore and the luciferase, thereby decreasing BRET. The protease activity is given by the ratio of the emission intensities of NLuc and the fluorophore, which can be measured by a digital camera or smartphone. We showed the superiority of small peptide tags in designing BRET-based biosensors, especially for those with multiple acceptors. This modular platform enables specific and highly sensitive detection of diverse proteases. By integrating any protease-specific cleavable peptide substrate, TePARs facilitate generalized protease sensing without laborious, enzyme-specific optimization, while maintaining high BRET efficiency and large dynamic range. This approach offers substantial promise for high-throughput drug screening and point-of-care diagnostics.

Results

Design and Optimization of MMP-9 Biosensor. As a first analyte, we chose matrix metalloproteinase-9 (MMP-9), a key protease involved in various inflammatory and cancer-related diseases. Currently, matrix metalloproteinase 9 (MMP-9) is measured by using a variety of analytical methods[27], however, the dependence on specific equipment and limited accessibility to end-users limit the practicality of these methods for quantitative point-of-care assays. We therefore aimed at generating a biosensor that measures MMP-9 in biological fluids through a ratiometric bioluminescent output.

To create a TePAR for MMP-9, we chose PRS↓LS as the peptide substrate (scissile bonds are identified by ↓). The selectivity of this sequence has been previously verified, confirming it as a specific substrate for MMP-9[28,28]. To generate a generalizable sensor, we fused the protease recognition sequence and luciferase or tetracysteine tag with a flexible GGTGGS linker respectively. Simple N- and C-terminal fusions of NLuc with a tetracysteine tag led to high intramolecular BRET (Fig. 1e) due to the small size of the label (12 amino acids). While this value is sufficiently high for a BRET biosensor used in point-of-care detection[22], we speculated that further improvement could be achieved by bringing the luciferase closer to the tetracysteine tag through circular permutation of NLuc (Fig. 1d)[29]. This increases the sensor's BRET ratio (the intensity of FlAsH emission relative to the intensity of NLuc emission) more than 50% to about 6.0 (Fig. 1e). Given that cpNLuc has also proven effective in fluorescent protein-based BRET sensor design, we compared TePARs to such analogs (Fig 1f). The FlAsH-labeled TePAR outperformed cpNLuc-YPet, an optimized yellow fluorescent protein with spectral properties similar to FlAsH, achieving a BRET ratio approximately 3-fold higher. For red-shifted sensors, cpNLuc-mCherry exhibited low BRET efficiency of about 10% due to limited spectral overlap with NanoLuc, whereas the ReAsH-labeled TePAR achieved approximately 70% efficiency, representing a 7-fold improvement.

To assess the ability of TePARs for reporting protease activity, we first incubated the sensor (cpNLuc-TC) with MMP-9 for a given time at room temperature. Protease cleavage separated the FlAsH-labeled tetracysteine tag from NLuc. This resulted in a significant increase in NLuc emission intensity and decrease in FlAsH emission intensity, leading to a substantial reduction in BRET efficiency from 86% to 12% (Fig. 1g).

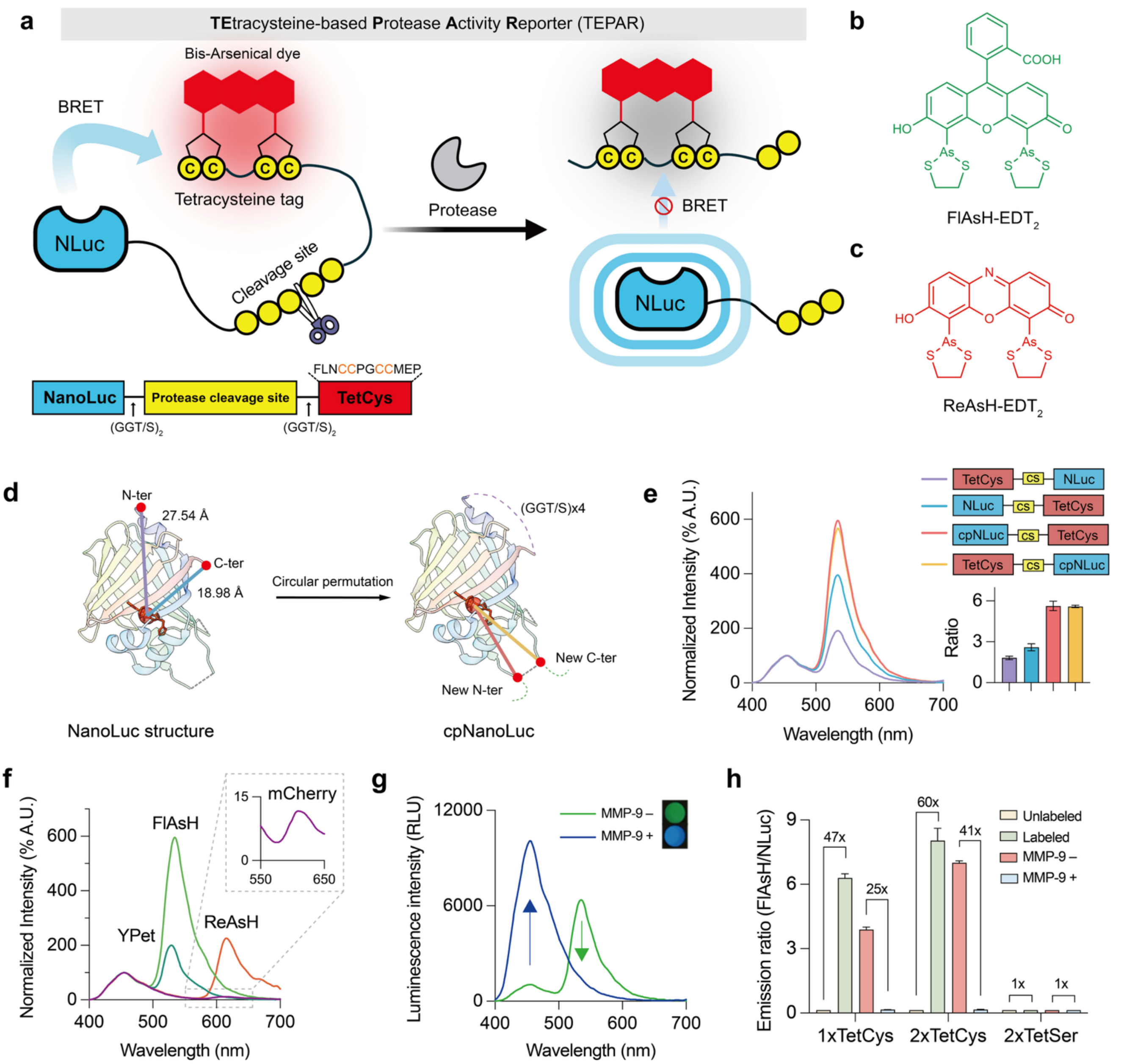


Fig.1 | Design and characterization of MMP-9 sensors. (**a**) Schematic of TePAR design. The fusion protein NLuc-protease cleavage sites-tetracysteine tag is conjugated to a bis-arsenical dye. Proteases can cleavage the fusion protein, reducing BRET efficiency. (**b**) (**c**) Chemical structure of the bis-arsenical dyes used to assemble the sensor. (**d**) 3-methoxy-furimazine-bound NLuc structure (PDB ID: 7SNT). Furimazine is displayed as a red stick structure. The distance between the active site (red ball) and the fluorophore attached at the protein terminus is reduced when using cpNLuc as the BRET donor, resulting in higher energy transfer efficiency. (**e**) Domain structures and emission spectra of TePAR variants. Positioning the tetracysteine motif at either N- or C-terminus of cpNLuc produced comparable BRET efficiencies, ~50% higher than a simple NLuc-TetCys fusion. (**f**) Emission spectra of MMP-9 sensor proteins labeled with FlAsH/ReAsH and cpNLuc-YPet/mCherry. TePARs showed superior BRET efficiencies for both green and red sensor compared to fluorescent protein-based sensors. (**g**) Spectra and color changes of the $TePAR_{MMP-9}$ in the presence or absence of MMP-9. MMP-9 cleavage induced a marked BRET reduction, resulting in a visible color shift detectable by a smartphone camera. (**h**) Emission ratios of cpNLuc-TetCys, 2xTetCys, and 2xTetSer in unlabeled and labeled states, and in the absence or presence of MMP-9. The serine mutant showed no background BRET from unbound dye, demonstrating that TePARs maintain minimal background signals.

An attractive feature of the sensor design is the small size of the tetracysteine tag, which allows for the incorporation of multiple identical acceptors to enhance BRET efficiency (Fig. 2a). Our initial rational designs, which included two and four tandem tetracysteine repeats connected by a flexible linker (GGSGG), yielded only marginal enhancements (c.a. 10%) in BRET ratio or even poorer performance (Fig. 2c). We hypothesized that these results could be attributed to neighboring fluorophore quenching and/or reduced labeling efficiency due to steric hindrance from the dense arrangement of multiple tetracysteine motifs. Similar observations have been reported in prior studies involving three tandem tetracysteine peptides, which showed only marginal improvements in total fluorescent intensity compared to a single tetracysteine motif[30]. Consequently, we explored the use of more rigid linkers, such as polyproline and α-helix (Fig. 2b), and found that the polyproline linker performed better. We further investigated proline linkers of varying lengths – 1, 3, and 5 amino acids – and discovered that a single proline linking two tetracysteine motifs yielded the highest BRET ratio for both FlAsH- and ReAsH-labeled sensor (Fig. 2c, Fig. S2). We suggest that the proline linker provides elasticity without the excessive rigidity of an α-helix, which might separate the tetracysteine motifs too far from NLuc, thereby improving labeling efficiency and reducing neighboring quenching but compromising overall BRET.

The optimized sensor, TePAR 2.0 (cpNLuc-2xTC), achieved ratios of 8.4 for FlAsH and 2.5 for ReAsH. The TePAR 2.0 displayed a 41-fold change in the emission ratio (FlAsH/NLuc) when titrated with MMP-9 (Fig. 2d). The c_{50} of the sensor (the MMP-9 concentration resulting in 50% maximum sensor response) was measured to be 31.5 ± 3.0 ng/mL after one hour incubation. From the initial slope of the titration curve, the LoD was determined to be as low as 1.9 ng/mL (20.9 pM). This LoD is 11-fold lower than that provided by the commercially available InnoZyme™ Gelatinase Activity Assay Kit based on a fluorescence resonance energy transfer (FRET) peptide[31]. Importantly, TePAR assay exhibits a wide dynamic range, spanning three orders of magnitude in concentration (3.2 ng/mL to 309.4 ng/mL), which eliminates the need for dilution when measuring biological samples.

We also assessed the selectivity of the $TePAR_{MMP-9}$ for MMP-9 over other proteins and small molecules, including glucose and other proteases that are active in blood (e.g., PSA and HCV NS3/4A protease). $TePAR_{MMP-9}$ showed high specificity for MMP-9 with minimal responses to other enzymes (Fig. 2e), including MMP-2, which cleaves certain FRET-based MMP-9 sensors[32,33]. This represents one important advantage in quantifying protease levels in biological fluids (e.g., plasma and serum), where the cross-reactivity with other active protease could compromise accuracy.

An important feature of the sensor is its large absolute BRET ratio change (~8 for FlAsH-labeled sensor and ~2.4 for ReAsH-labeled sensor) and rapid reaction kinetics. We attribute this large change to (i) the circular permutated NLuc, which brings the luciferase active site in close proximity to the fluorophore, and (ii) the incorporation of multiple acceptors, which increases BRET efficiency up to 90%. Furthermore, the kinetics of TePAR are faster than the conventional FP-based sensor. This rapid kinetics can be attributed to the small size of the tetracysteine tag, which effectively minimizes steric hindrance (Fig. S3). To confirm this, we measured the kinetic profile of TePARs and the fluorescent protein-based sensor cpNLuc-YPet. TePARs showed faster response, with a half-life ($t_{1/2}$) of 5-6 min, and the additional tetracysteine tag did not largely impair kinetics (Fig. 2f, S4). We also compared the kinetic properties of them. TePAR 2.0 achieved markedly higher catalytic efficiency, with a V_{max} value 3.2-fold greater than that of cpNLuc-YPet and a k_{cat}/K_M ratio of 7.31×10^5 M^{-1} s^{-1} (Fig. 2g, h), representing more than 4.4-fold improvement over cpNLuc-YPet. These results underscore the exceptional sensitivity of TePAR, which achieves an unprecedented LoD for MMP-9 of 20.9 pM. Moreover, due to the fluorogenic properties of biarsenical dyes, the background BRET is negligible (Fig. 1h, S5). After complete cleavage of the sensor protein, the background BRET ratio remains low, endowing TePARs with a high signal-to-noise ratio.

We performed all the above measurements using a traditional microplate reader, which is not suitable for point-of-care applications. Next, we tested whether the color change of bioluminescent signal could be detected by a smartphone-camera (Fig. S6). We placed the microplate into a simple polystyrene icebox to exclude ambient light and took a picture with a smartphone camera through a hole in the box. Ratios of the blue to green channels were plotted, and a fitting curve was calculated to derive an equation for estimating MMP-9 concentration from the emission color. The results from microplate reader and those calculated from the smartphone camera images showed a high correlation (Pearson's r = 0.99). Therefore, smartphone camera-based measurements of protease activity strongly agree with the results from specialized instrumentation, demonstrating its potential as an alternative method. Such visual readouts are particularly valuable for resource-limited or point-of-care settings, where speed and accessibility often outweigh sensitivity.

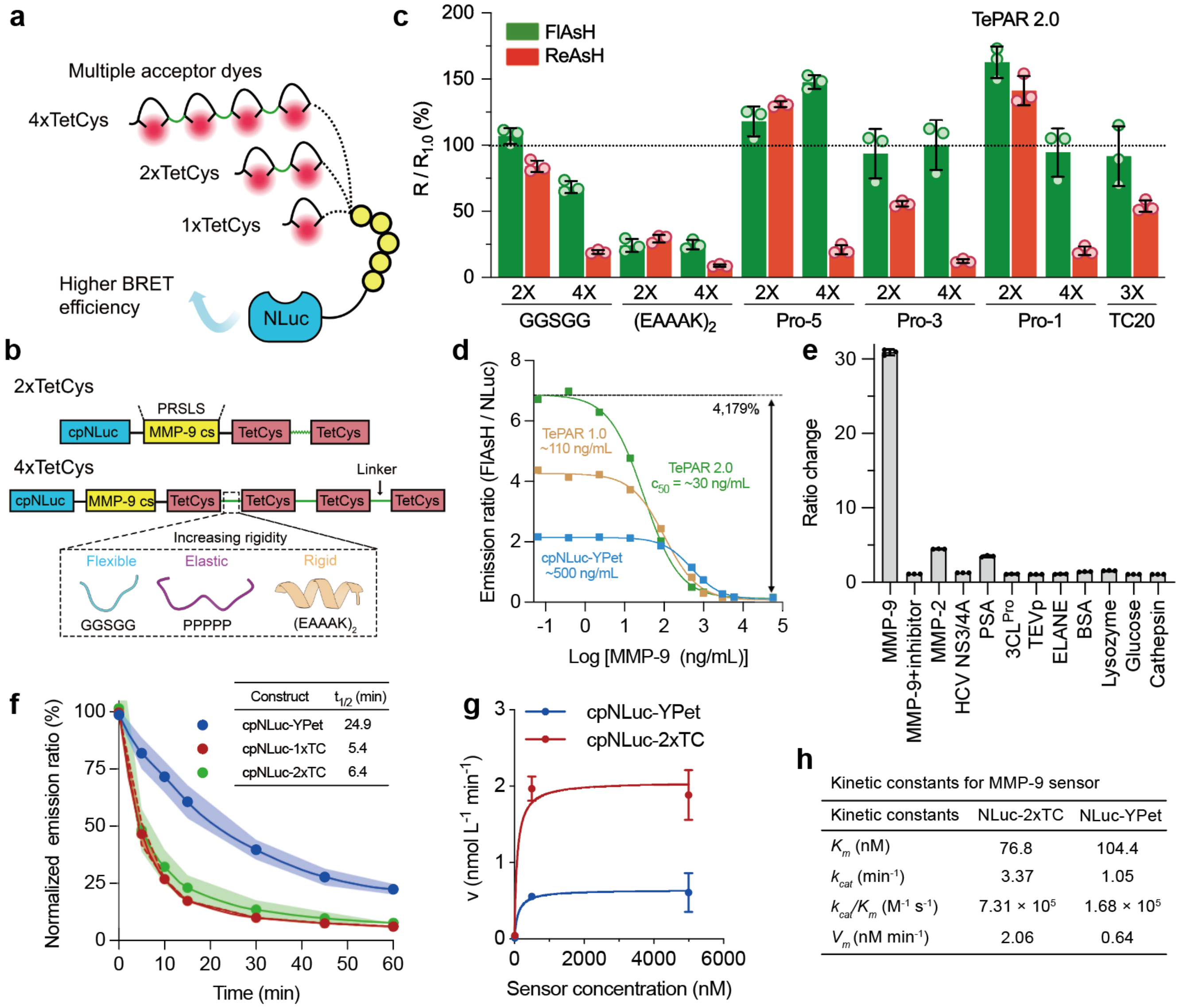


Fig. 2 | Sensor optimization. (**a**) The optimizing strategy used to increase BRET efficiency. Multiple TC tags are connected through designated linkers to enhance energy transfer efficiency. (**b**) The linear diagrams of the sensor protein variants. The linker structures were generated by Alphafold2[34]. (**c**) Relative BRET ratio ($R/R_{1.0}$) for each linker composition and number of TC tag is plotted based on the bioluminescence signal of 10 nM sensor protein in HEPES buffer. The optimal combination was a single proline to link two TC motifs. This molecule is named TePAR 2.0. (**d**) Titration curves of sensor variants. Emission ratios (FlAsH/NLuc) were plotted as a function of MMP-9 concentrations (obtained from three independent titrations; the graph shows one titration). TePAR 2.0 exhibited a c_{50} ~17-fold lower than cpNLuc-YPet, indicating superior sensitivity and dynamic range. (**e**) BRET ratio changes of TePAR 2.0 (10 nM) after incubation with various protease and controls at 37 °C for 60 min. TePAR 2.0 showed robust response to MMP-9 but negligible changes with other analytes. MMP-9, TEV protease, ELANE, cathepsin: 1 μg/mL. MMP-9 inhibitor (Ilomastat): 100 nM. MMP-2, thrombin, HCV, PSA, $3CL^{pro}$: 0.5 μg/mL. BSA: 10 mg/mL. lysozyme: 2 μg/mL. Glucose: 10 mM. Data as means ± SD ($n = 3$). (**f**) Proteolysis kinetics of TePARs and cpNLuc-YPet in the presence of MMP-9. Sensor proteins at 10 nM were incubated with 500 ng/mL MMP-9 at 37 °C. The average and standard deviations are shown for $n = 3$. (**g**) Cleavage rate of sensor proteins by MMP-9 (50 ng/mL) as a function of sensor concentration. Data as individual replicates and kinetic values determined using the Michaelis-Menten equation ($n = 3$). (**h**) Kinetic constants for NLuc-2xTC and NLuc-YPet.

Engineering TePARs for Different Proteases. To demonstrate the modularity of our approach, we expanded our repertoire of TePARs to include four commonly used protease biomarkers for disease diagnosis: MMP-2, which belong to the metalloproteinase family; thrombin and prostate-specific antigen (PSA), which are serine proteases; and hepatitis C virus (HCV) NS3/4A protease, which is a cysteine protease. Due to the simple design and easy production of TePARs, We can generate a new protease sensor with robust performance within a week: (1) design of the protease-specific cleavage sequence and construction of the TePAR plasmid; (2) transformation of the plasmid into *E. coli* for induced expression of the sensor protein; (3) purification and labeling to assemble functional sensor; and (4) potential applications, including high-throughput drug screening, point-of-care testing assays and so on (Fig 3a, S7). It is noteworthy that the exceptional performance of most TePARs was achieved through a straightforward substitution of the proteolysis-responsive peptide sequence (Table S3) without any complex geometry optimization[20–22], thus underlining the generality of the TePAR platform to transform peptide substrate into bioluminescent sensors.

MMP-2 is a potential predictive marker for papillary thyroid carcinoma and chronic obstructive pulmonary disease (COPD). Measuring MMP-2 levels in plasma can predict the clinical progression of Chagas disease and serves as a diagnostic marker[35]. The resulting $TePAR_{MMP-2}$ have a c_{50} of 25 ng/mL, which aligns well with clinically relevant concentrations (Fig. 3e). Thrombin is a key protease involved in blood coagulation, catalyzing the conversion of fibrinogen to insoluble fibrin, which then polymerizes to form a stable clot. Its cleavage activity can be an important index of coagulation status. Our titration results showed that $TePAR_{Thrombin}$ exhibited a remarkable ratio change of 2,251.1%, c_{50} of 32.0 ng/mL and LoD of 3.8 ng/mL[22] (Fig. 3f), surpassing the Thrombastor biosensor (LoD 10 nM or 370 ng/mL). This performance is competitive with existing small-molecule- and protein-based thrombin sensors (Table S4). PSA is an FDA-approved biomarker for prostate cancer diagnosis and screening. Quantifying PSA levels and its serum isoforms helps distinguish between prostate carcinoma and benign inflammatory conditions[36]. We screened three peptide substrates[37] and selected Arg-Arg-Leu-Asn-Tyr-Ser-Leu for its rapid cleavage and the highest BRET ratio in TePAR construct (Fig. S8). The resulting $TePAR_{PSA}$ show LoD 88.7 ng/mL (Fig. 3g). Given the inherently low enzymatic activity, sample preconcentration may enhance its utility for clinical ranges. Finally, The NS3/4A protease from HCV plays a crucial role in viral replication. The NS3 serine protease requires the accessory viral protein NS4A for optimal cleavage activity. NS3/4A can be used to assess HCV infection and serves as a valuable target for developing new HCV antivirals[38]. Notably, despite the cleavage site being 20 amino acids long, the BRET ratio of the $TePAR_{NS3/4A}$ remains high, reaching 3 for FlAsH-labeled sensor and 0.3 for ReAsH-labeled sensor (Fig. 3h), which is sufficient for camera detection. The ΔR_{max}, c_{50}, response range and LoD of all the TePARs generated in this work are summarize in Table 1. These results demonstrate the promising applicability of TePARs for point-of-care detection of various protease biomarkers.

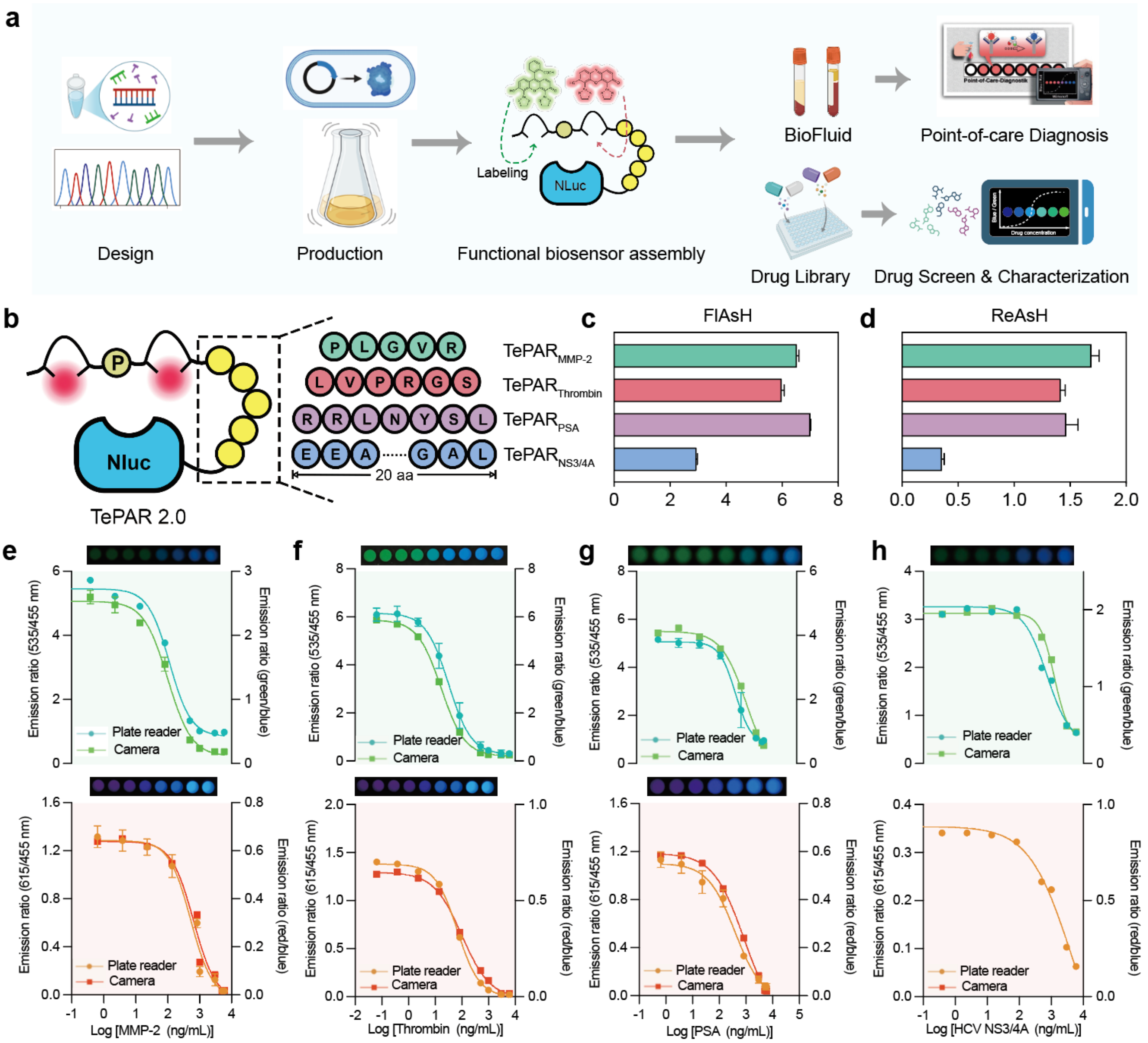


Fig. 3 | Modularity in the analysis of different proteases. (**a**) Flowchart illustrating TePAR generation. Cleavage sequence design and plasmid construction; *E. coli* transformation for expression; purification/labeling; and applications like drug screening and point-of-care testing. This enables rapid (1-week) sensor creation for new targets. (**b**) Cleavable peptide sequences of MMP-2, thrombin, PSA, HCV NS3/4A protease. Simple swapping of these validated sequences produced functional TePARs for each target without additional optimization. **c-d**, BRET ratios of FlAsH-labeled (**c**) and ReAsH-labeled (**d**) TePARs for different proteases remained consistently high without further optimization. **e–h**, Dose–response curves for MMP-2 (**e**), thrombin (**f**), PSA (**g**) or HCV NS3/4A protease (**h**). Top, FlAsH-labeled sensors. Bottom, ReAsH-labeled sensors. Most sensors exhibited wide dynamic ranges, low LoDs, and large ratio changes, suitable for potential clinical applications. Data are mean ± s.d. Images were captured using a smartphone camera.

Table 1. The ΔR_{max}, c_{50} [ng/mL], response range [ng/mL] and LoD [ng/mL] of the TePARs for the five proteases.

Sensor	ΔR_{max} [%]	C_{50}	Response Range	LoD
MMP-9	6718.9	31.5 ± 3.0	3.2 – 309.4	1.91
MMP-2	2244.1	102.8 ± 14.2	15.7 – 672.2	37.7
Thrombin	2251.1	32.0 ± 0.7	3.7 − 279.2	3.8
PSA	656.6	514.2 ± 28.2	110.6 – 2390.4	88.7
NS3/4A	803.3	730.5 ± 208.3	114.4 – 4664.7	25.9

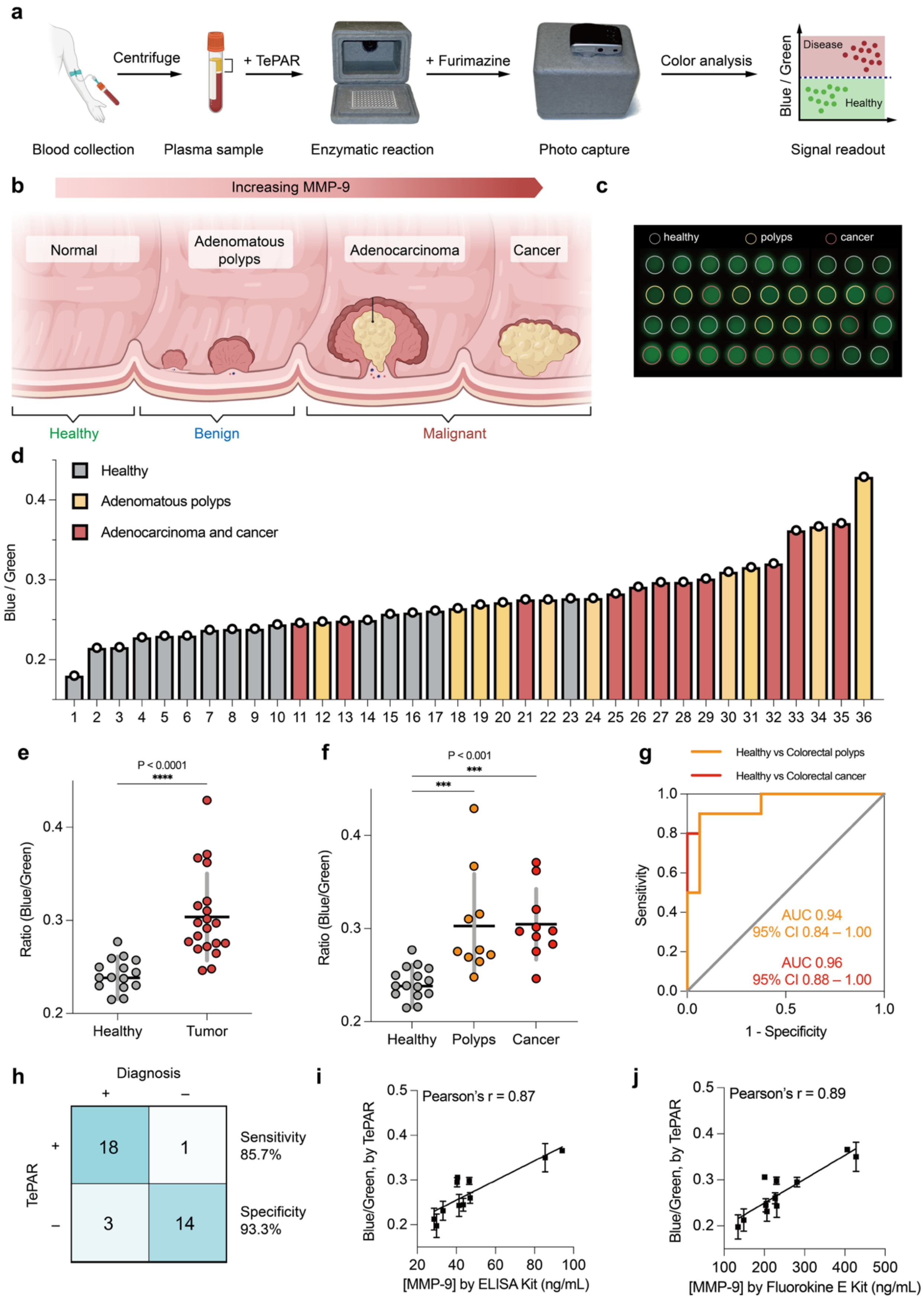
TePAR for Point-of-Care Diagnosis in Colorectal Disease Patients. MMP-9 is a critical enzyme involved in tumor invasion, metastasis, and angiogenesis. Its activities is directly correlated with tumor growth and invasion, making it a novel biomarker and potential therapeutic target in human cancers[39]. Therefore, monitoring the levels of MMP-9 is essential for the effective treatment and prognosis of cancer patients. Plasma levels of MMP-9 have been analyzed in various cancers, including colorectal cancer, which typically begins as adenomatous polyps, with only 5-10% of these polyps progressing to malignant cancer (Fig. 4b). Current methods for MMP-9 detection are limited by prolonged assay time, suboptimal sensitivity, high cost and dependence on specialized laboratory instrumentation, rendering them unsuitable for point-of-care applications (Table S5). A rapid, accurate detection method requiring minimal sample handling would significantly enhance diagnostic capabilities of colorectal disease.

To achieve point-of-care measurement of MMP-9, we developed an assay where a drop of plasma is diluted in a solution containing the TePAR and analyzed using a smartphone camera mounted in a simple holder (Fig. 4a). MMP-9 levels are indicated as the blue/green emission ratios, and the entire measurement process takes less than 60 minutes, requiring only 25 μl of plasma or less. The ratiometric nature of the TePARs enhances their applicability in point-of-care settings, as variations in sensor protein concentration or the volume of the plasma drop do not significantly affect the results.

We assessed the clinical performance of the TePAR assay through side-by-side tests comparing the TePAR assay results, clinical diagnoses by enterologists and readings from both an ELISA kit and an MMP-9 activity kit, using 36 clinical plasma samples (patients with adenomatous polyps, n = 10; patients with adenocarcinoma or colorectal cancer, n = 11; and healthy controls, n = 15). Plasma samples were collected for a double-blind test followed by clinical validation. For the diagnosis of colorectal tumor, endoscopy and biopsy were performed by enterologists for each patient in the hospital (Fig. S9). Demographic characteristics of participants are presented in Table S6. In the translational study, we directly used the blue/green ratio values from the TePAR assay of clinical plasma samples and did not convert them to corresponding MMP-9 activity values (Fig. 4d, S9). The mean ratios of plasma samples of patients with colorectal tumor (blue/green = 0.30 ± 0.03) were significantly higher than those of healthy individuals (blue/green = 0.24 ± 0.02, $P < 0.0001$) (Fig. 4e). Notably, the TePAR assay could distinguished patients with polyps, which are at the early stages of cancer development, from healthy controls ($P < 0.001$) (Fig. 4f). Next, we plotted the receiver operating characteristic (ROC) curve of the TePAR assay and calculated the area under the curve (AUC) to assess its diagnostic accuracy (Fig. 4g). The estimated AUC values of patients with colorectal polyps and cancer versus those for healthy individuals were 0.94 and 0.96, respectively, indicating similar accuracy between two patient groups. The sensitivity and specificity of the TePAR assay for patients with colorectal tumor against those healthy controls were 85.7% and 93.3%, respectively, at the cut-off value of 0.2611 (Fig. 4h).

Furthermore, TePAR assay results correlated strongly with those from commercial kits: the Human MMP-9 ELISA Kit (Abcam), which employs capture antibodies conjugated to an affinity tag recognized by plate-coated monoclonal antibodies for single-step sandwich complex formation, and the Human Active MMP-9 Fluorokine E Kit (R&D Systems), which uses a monoclonal antibody for MMP-9-specific capture and a fluorogenic peptide substrate for activity readout. Analysis of the patient plasma samples revealed Pearson correlation coefficients of 0.87 versus ELISA kit (Fig. 4i) and 0.91 versus activity kit (Fig. 4j). The assay also exhibited excellent reproducibility, with an average coefficient of variation of 7% for technical triplicates (Fig. S10). It is also instructive to compare the performance of our improved TePARs with conventional fluorescent protein-based sensors, we observed that TePARs demonstrated a 189% ratio change when measuring plasma samples of patient 1 and 36, while the fluorescent protein-based sensor showed only a 2% increase over background (Fig. S11).

The simplicity of this assay protocol suggests the potential for patient self-testing, although further modifications are necessary. Currently, the assay requires a 60-minute incubation period, however, this could be reduced for patients with advanced malignant tumor. Additionally, transitioning to a more automated assay format based on microfluidic devices could streamline the process. Addressing these technical challenges could provide a valuable tool for cancer patients, facilitating timely monitoring of protease activity. Moreover, our activity assay has the potential of scalability test of up to 384 samples with one camera shot. Among the assay designed for point-of-care use, with a focus on clinical relevance in conditions including cancer, the TePAR assays stands out for its high sensitivity and specificity, all achieved at an extraordinarily low cost per assay (Table S7).


a
Centrifuge
+ TePAR
+ Furimazine
Color analysis
Blood collection
Plasma sample
Enzymatic reaction
Photo capture
Signal readout
Blue / Green
Disease
Healthy
b
Increasing MMP-9
Normal
Adenomatous polyps
Adenocarcinoma
Cancer
Healthy
Benign
Malignant
c
healthy
polyps
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Healthy
Adenomatous polyps
Adenocarcinoma and cancer
Blue / Green
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P < 0.0001
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Ratio (Blue/Green)
Healthy
Tumor
f
P < 0.001
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Ratio (Blue/Green)
Healthy
Polyps
Cancer
g
Healthy vs Colorectal polyps
Healthy vs Colorectal cancer
Sensitivity
1 - Specificity
AUC 0.94
95% CI 0.84 – 1.00
AUC 0.96
95% CI 0.88 – 1.00
h
Diagnosis
TePAR
18
1
3
14
Sensitivity 85.7%
Specificity 93.3%
i
Pearson's r = 0.87
Blue/Green, by TePAR
[MMP-9] by ELISA Kit (ng/mL)
j
Pearson's r = 0.89
Blue/Green, by TePAR
[MMP-9] by Fluorokine E Kit (ng/mL)

Fig. 4 | Point-of-care detection of MMP-9 in clinical samples. (**a**) Flow diagram of the TePAR-based point-of-care assay with smartphone camera readout. Blood samples are drawn from patients and centrifuged to obtain plasma; 25 μL plasma is mixed with assay buffer containing TePAR to initiate the enzymatic reaction; a photo is captured via smartphone, and the emission ratio is analyzed for quantitative readout. (**b**) Schematic illustrating the progression of colorectal cancer, accompanied by elevated MMP-9 levels. (**c**) Images. (**d**) Blue/green ratios of the patients with adenomatous polyps (abbreviated as "polyps", $n = 10$), adenocarcinoma or colorectal cancer (abbreviated as "cancer", $n = 11$), and healthy controls ($n = 15$). **e**, **f**, Blue/green ratios of plasma samples acquired from TePAR assay of patients with colorectal tumor ($n = 21$) (**e**) and patients with polyps ($n = 10$), cancer ($n = 11$) (**f**) versus healthy controls ($n = 15$ clinical plasma samples). The lines represent the mean, and the error bars represent s.d. Two-tailed Student's *t*-tests were used for comparisons. ****$P < 0.0001$. ***$P < 0.001$. (**g**) ROC analysis for the discrimination between colorectal patients and healthy individuals using TePAR. AUC is defined as the area under the ROC curve. (**h**) Concordance of the clinical diagnostic results obtained using TePAR for 36 patient plasma samples. The cut-off value from the ROC curve to determine the sensitivity and specificity of TePAR was 0.25. **i**, **j**, Correlation of MMP-9 levels in 11 patient plasma samples quantified by TePAR versus the Human MMP-9 ELISA kit (**i**) and Human Active MMP-9 Fluorokine E Kit (**j**).

TePAR for MMP Inhibitor Screening and Drug Characterization Assays. We next assessed the utility of the TePAR in MMP inhibitor screening and drug characterization assays. First, we adapted our $TePAR_{MMP-2}$ and an MMP inhibitor library (Table S8) to a 96-well plate format for camera-based assays that could screen small molecules for their ability to inhibit MMP-2 activity (Fig. 5a). We used a simple smartphone camera to test a collection of anticancer and anti-inflammatory drugs with varying MMP-2 inhibition efficacy. In this assay, $TeAPR_{MMP-2}$ was incubated with recombinant human MMP-2 protein and each individual drug for 2 hours, followed by addition of furimazine and capture of bioluminescent images (Fig. 5b). Green/blue emission ratios served as a metric to compare the inhibitory potencies of all 36 drugs. (Fig. 5c). Among these, the broad spectrum MMP inhibitor GI254023X and selective MMP-2 inhibitor Ilomastat exhibited the highest ratios and were selected for further studies. The compounds exhibiting the weakest inhibitory potency against MMP-2 was identified as doxycycline, an antibiotic with a high IC_{50} 56 mM[40]. This data aligns with our experimental observation of visible blue emission in the assay well (A2), indicative of uninhibited protease activity.

Additionally, we developed a one-pot assay that combined the drug, MMP-2, TePAR and furimazine in a single well, allowing for the continuous tracking of color changes. The bioluminescence signal lasted for > 90 min to enable long-term monitoring of inhibitory kinetics. Using this assay, we assessed Ilomastat, GI254023X, and ARP-100 (a widely used selective MMP-2 inhibitor), quantifying blue/green emission ratios over 90 min across varying drug concentrations (Fig. 5e). From endpoint images at 90 min (Fig. 5f), we generated ratio–concentration curves (Fig. 5g) and calculated half-maximal inhibitory concentrations (IC_{50} values) to evaluate potency. These IC_{50} values aligned with literature reports, demonstrating that the simplicity of our camera-based assay does not compromise accuracy (Fig. 5h). The close agreement confirms the platform's reliability and validates its equivalence to conventional assays for drug discovery applications. These findings highlight TePARs' potential as a low-cost, camera-based platform for protease inhibitor discovery, obviating the need for expensive and specialized plate readers and thereby accelerating drug screening while enhancing accessibility in resource-limited settings.

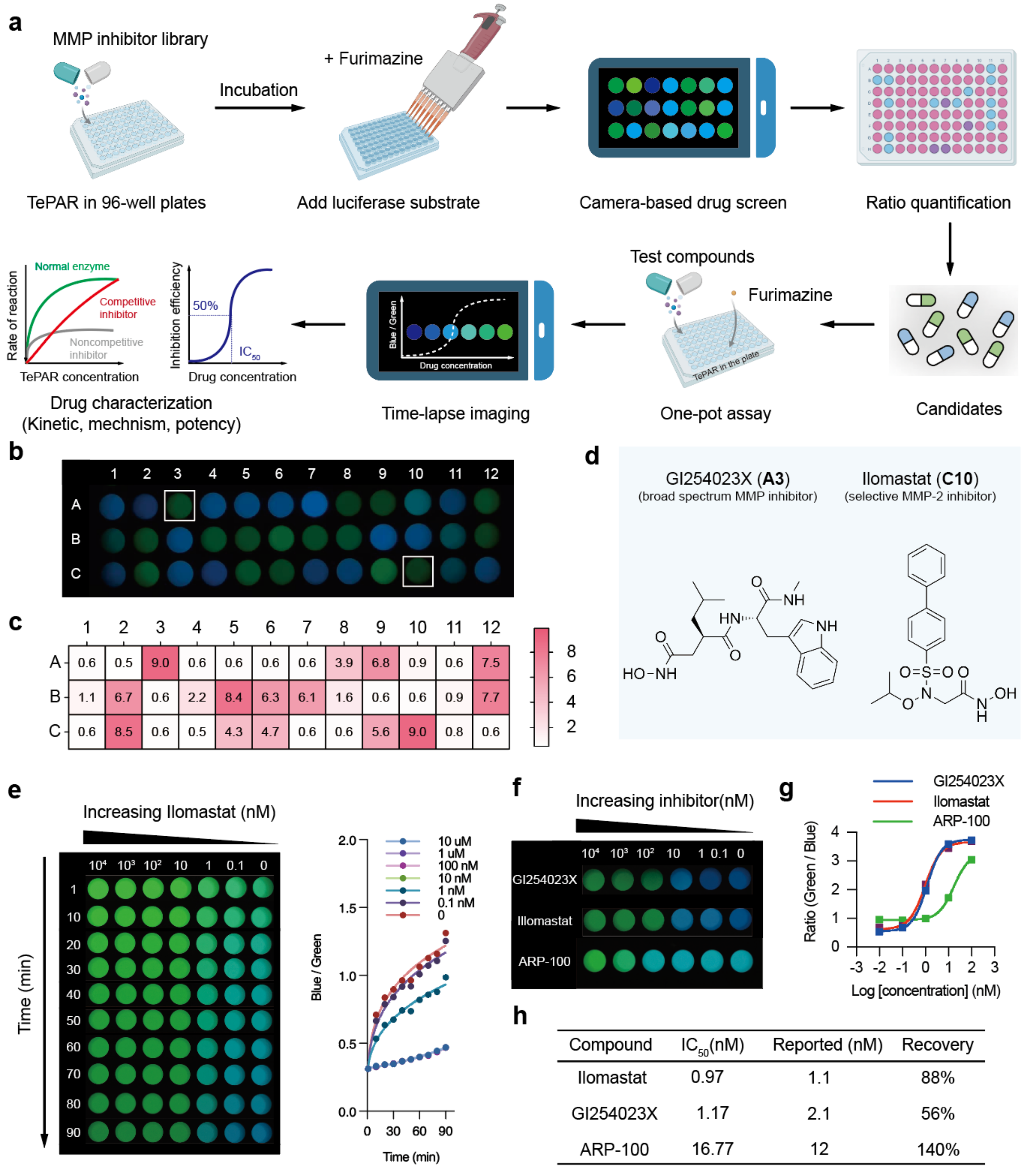

Fig. 5 | High-throughput screening of MMP inhibitors using TePAR. (**a**) Schematic of the experimental protocol for the TePAR-based drug screening assay. The drug library, protease, and TePAR are incubated for a defined period, followed by furimazine addition and smartphone camera imaging to capture bioluminescent signals. Images are quantified via emission ratios to screen potent candidates. Leveraging the sustained bioluminescence, furimazine can alternatively be incorporated with test compounds in a one-pot format. This enables time-lapse imaging to monitor inhibitory kinetics, elucidating drug mechanisms and potency. (**b**) Image of $TePAR_{MMP-2}$ incubated with a drug library for 90 minutes. (**c**) Heatmap of green/blue ratios. Higher green percentage indicates greater inhibitor potency of the drug. (**d**) Chemical structures of the top two drugs showing highest MMP-2 inhibitory potency. (**e**) Left, real-time monitoring images of the proteolysis reaction with varying Ilomastat concentration (ranging from 0.1 nM to 10 μM) over 90 minutes. Right, quantified emission ratios from the images. (**f**) Images of $TePAR_{MMP-2}$ incubated with selected drugs at varying concentrations for 90 minutes. (**g**) Dose-response curves derived from imaging data. (**h**) IC_{50} values calculated from images and comparisons with literature data. The IC_{50} values determined by TePAR closely align with those reported in the literature.

Discussion

TePARs meet most of the ASSURED criteria set by the WHO to qualify as efficient diagnostic testing: Affordable, Sensitive, Specific, User-friendly, Rapid and robust, Equipment-free and Deliverable to end-users, making them promising candidates for protease detection. One of the key parameters of any ratiometric biosensor is its maximum ratio change. TePARs can quantify proteases with significantly high ratio changes, comparable to sensor systems that have successfully applied for point-of-care quantification of phenylalanine, glucose, and glutamate[19]. This capability clearly surpasses the ratio changes of any previously reported BRET-based indicators.

We have demonstrated the versatility and sensitivity of the TePAR platform for point-of-care protease detection. TePARs represent a significant advancement toward a reagent class that can be produced and utilized with relatively low-cost infrastructure. Despite the advantages TePARs offer over antibody-based detection methods, several steps still need to be taken to transition them from proof-of-concept to practical devices. First, Stability optimization: The stability of the sensor protein and luciferase substrate must be evaluated and potentially optimized. The cysteine-rich sensor protein is prone to crosslinking via disulfide bridges, which impairs labeling efficiency. This issue can be mitigated by storing the protein in a TCEP-containing buffer or pre-labeling reduction with TCEP, though TCEP may disrupt protein structure near cysteine residues[41]. Therefore, cysteine-free self-labeling peptide tags[42–44] is essential for consistent and reliable performance. Additionally, brighter dyes could improve camera-based sensitivity, and improved labeling efficiency and kinetics could shorten assay times. Second, Sensitivity for low-activity proteases: For proteases with inherently low activity or those inhibited in plasma by alpha-2-macroglobulin (α_2M), the activity-based TePARs may lack sufficient sensitivity due to slow enzymatic kinetics and intrinsically low concentrations. To address this, we need sample pre-processing methods to minimize environmental interferences or leveraging liquid-liquid phase separation to accelerate enzymatic rates[21].

In conclusion, TePARs are a versatile family of chemigenetic, bioluminescent sensor proteins that can be engineered to detect a wide range of proteases. These sensors facilitate the quantitative analysis of patient samples and high-throughput drug screening using simple, portable devices. By making protein activity monitoring more accessible, TePARs have the potential to enhance early disease detection, personalized medicine, and therapeutic interventions.

Methods

General information. All reagents were purchased from commercial suppliers and used without further purification. Detailed information on the reagents and resource is listed in Supplementary Information.

Sensor construct. All sensor constructs were ordered from Genscript Biotech and were subcloned into a pET backbone (https://www.addgene.org/187106/). The amino acid sequences of the described sensor proteins can be found in Supplementary Information.

Sensor protein expression and purification. Plasmids encoding our sensors were used to transform *E. coli* BL21 (DE3) competent cells (Tsingke). Transformants from a single colony were used to inoculate 10 mL of LB supplemented with 100 μg/mL ampicillin (LB-Amp) and grown overnight at 37 °C, 180 rpm. The overnight seed culture was diluted 1:100 LB-Amp medium and was incubated at 37 °C, 180 rpm, until the OD_{600} was between 0.4 and 0.6. Then the culture was cooled down to 16 °C and isopropyl β-d-thiogalactopyranoside (IPTG) was added (0.5 mM) to induce overnight protein expression. After induction, cells were harvested by centrifugation at 4,000 *g* for 30 min and lysed by sonication in the presence of 1 μg/mL lysozyme and 1 mM protease inhibitor cocktail (Meilunbio). The cellular debris was removed with centrifugation (12000*g*, 30 min, 4 °C), and the supernatant was cleared using a 0.45 μm syringe filter (BioLand). The filtered lysate was sequentially loaded onto Ni-NTA (Qiagen) and Strep-Tactin (IBA) columns for purification. The purified protein was concentrated using a centrifugal filter device (Amicon Ultra-0.5) with a cut-off of 10 kDa and were stored in storage buffer at -20 °C until further use. The protein concentration was determined using a Bradford assay (ThermoFisher Scientific) with BSA as a standard. Typical protein yields were between 1 to 5 mg/L of bacterial culture.

Biosensor labeling. Sensor proteins were treated with 2 mM tris(2-carboxyethyl)phosphine (TCEP) overnight at room temperature to ensure cysteines that bind to dyes are completely reduce before labeling. For tetracysteine tag labeling, the sensor proteins were diluted to concentration of 1 μM in MOPS buffer containing 15 μM of FlAsH-EDT_2 or ReAsH-EDT_2. The resulting sensors were used without further purification. Although monothiols such as 2-mercaptoethanol were helpful to catalyze the labeling process, caution is advised when considering its application for metalloproteinase sensing due to its potential inhibitory effects[45].

Emission spectra and titration curves. The labeled sensors were diluted to concentrations of 10 nM in 50 μl TCNB buffer spiked with known concentrations of analyte in white nonbinding 96-well plates (Corning). After incubation at 37 °C for 60 min, 50 μl furimazine (20 μM) in HEPES buffer was added. Spectra were measured on a microplate reader (BioTek Synergy H1, Agilent) with a step size of 5 nm, a bandwidth of 10 nm and an integration time of 50 ms. The ratio of donor and acceptor emission was plotted against the analyte concentration, and the data were fitted to a four-parameter logistic (4PL) model using nonlinear regression analysis to obtain the c_{50} and maximum ratio change.

Screening of protein variants using bioluminescent readout. The screening aims to identify the sensor protein with the highest BRET ratio. Labeled proteins were diluted to 10 nM in 50 μl HEPES buffer. To obtain bioluminescent readout, 50 μl furimazine (20 μM) was added. The bioluminescent emission spectrum for each protein variant was then recorded following the emission spectrum protocols.

Protease kinetics calculations. Sensor proteins were combined at different concentrations with MMP-9 in TCNB buffer at 37 °C. The reactions were then analyzed at various timepoints by measuring emission spectrum. The obtained ratios were normalized to a blank reference to determine the percentage of uncleaved substrate. We calculated initial reaction velocities at substrate consumption <25%. Data were fit to a Michaelis–Menten enzyme-kinetics model, with center values representing the mean and error bars representing the standard deviation of three technical replicates.

TePARs for smartphone-based assay. Solutions in 96-well plates were prepared the same way as described above for titration curves. Briefly, 25 μL of protease at varying concentrations was mixed with 25 μL of 20 nM labeled sensor protein, yielding a final sensor concentration of 10 nM across all protease analytes. Following incubation, 50 μL of 20 μM furimazine in HEPES buffer was spiked and a picture of the 96-well plate was taken using a smartphone camera through a hole in a polystyrene ice box to prevent light from the environment to disturb the measurement. The following smartphone camera settings were used: focal length = 27 mm, ISO = 1000, exposure time = 1-2 s. Acquired images were processed with ImageJ software to detect luminescent spots and quantify average intensities for blue and green or red channels from the RGB color model.

Clinical trial. The study included a total number of 36 subjects, with detailed demographics provided in Supplementary Information. Subjects were diagnosed between November and December 2024 at Zhongshan Hospital of Fudan University (Shanghai, China). Clinical diagnoses were based on physical examinations, blood tests, biopsies and endoscopic evaluations. Subjects with a history of heart attack or heart failure were excluded from the study. Blood samples were collected from all patients prior to surgical treatment, with none having received chemotherapy or radiotherapy before sample collection. To standardize clotting conditions, plasma was separated within 1 hour of collection, aliquoted, and stored at -80 °C until analysis. Plasma levels of MMP-9 were measured using the Human MMP-9 ELISA Kit and Human Active MMP-9 Fluorokine E Kit according to the manufacturer’s instructions, with samples diluted 100-fold prior to assay. For the TePAR assay, 25 μL of a stock solution containing 20 nM labeled sensor was added to 25 μL of patient samples. After a 60-minute incubation at 37 °C, 50 μL of 20 μM furimazine diluted in HEPES buffer was added, and detection was performed using a digital camera as previously described.

Drug screening assay. For high-throughput screening, 20 nM $TePAR_{MMP-2}$ was incubated with 2 µg/mL MMP-2 and a drug library with each compound at 1 nM for 90 min at 37°C in a total volume of 50 µL. Subsequently, 50 µL of 20 µM furimazine was added, and bioluminescence images were then acquired using the camera-based assay described previously. For kinetic and IC_{50} measurements, furimazine was added with other components at first and time-lapse imaging was performed for 90 minutes.

Statistics and reproducibility. Statistical significance was assessed through unpaired two-tailed t tests. All collected data were included in the analyses without exclusion. The following notations apply for all statistical analyses: NS (not significant), $P \geqslant 0.05$, $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$ and $^{****}P < 0.0001$.

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